

Name: _____

Exam Instructions

- This is a closed book exam. No notes or books are allowed.
- You may use a calculator, but it won't be necessary.
- WRITE LEGIBLY. Illegible answers will not be graded.
- Preferably, use the provided solution boxes for your answers. If you need more space, make sure to clearly indicate which question you are answering.
- You have 2 hours to complete the exam.
- The final score will be graded out of 100, with a maximum of 110 points available (the additional 10 points are meant to give you flexibility, you can make minor mistakes and still obtain a perfect grade)
- You can use the list of identities and properties at the end of the exam.

1. **Bayesian Inference.** In a simplified model of weather, we assume that each day is either *sunny* or *not sunny*. Let x represent how many out of n consecutive days of observations it was sunny, i.e., if $x = 3$ and $n = 7$, then it was sunny for 3 out of 7 days. We denote the unknown probability of a sunny day by $\theta \in [0, 1]$ and we want to model this problem using a Bayesian approach (*Hint*: remember a similar example from the lectures).

- (3) (a) Given data x and parameter θ , write down the Bayes' theorem for the posterior distribution of θ given x . Describe the meaning of each term in the equation.

- (2) (b) What is the primary goal of Bayesian inference?
- ☐ To select hyperparameters via cross-validation
 - ☐ To provide a density function for the parameters conditioned on the observed data
 - ☐ To enforce sparsity in the parameter estimates
 - ☐ To transform non-linear models into linear ones through kernel methods

- (5) (c) Identify the appropriate probability model for x given θ . Then, choose a distribution for θ and justify your choices in both cases.

- (5) (d) We now want to compute the probability of sunny θ given the observations x . Discuss the steps you would take to do it. What is this probability? How would you compute it? Is there a property that would make this computation easier?

- (7) (e) Derive this probability. (*Hint*: You can use the result from the previous part.)

- (5) (f) Consider again the weather model where each day is either sunny or not, and let $\theta \in [0, 1]$ denote the unknown probability of a sunny day. Suppose the true value of θ is 0.5. As we observe more days ($n \rightarrow \infty$), which of the following statements is true about the posterior distribution of θ ?
- ☐ The posterior distribution becomes increasingly concentrated around 0.5, regardless of the initial prior, and approaches a normal distribution.
 - ☐ The posterior distribution remains centered at the prior mean and does not change significantly with more data.
 - ☐ The influence of the prior becomes stronger as more data is observed.
 - ☐ The posterior variance increases with more observations, indicating greater uncertainty.
- (5) (g) Assume we have M different models for the data, $\mathcal{M}_1, \dots, \mathcal{M}_M$. What tool can we use to select the best model? Describe the steps you would take to do it.

2. **Generalized linear regression.** Suppose we collected the number of customer arrivals $y_1, \dots, y_n \in \mathbb{N}^+$ at a shop during n different days, together with some known covariates $\mathbf{x}_1, \dots, \mathbf{x}_n \in \mathbb{R}^d$ representing features such as weather, promotions, weekday indicators and other factors. Our objective is to model the number of arrivals as a function of these features as a regression problem. We assume a generalized linear model with a Poisson likelihood:

$$y_i \mid \mathbf{x}_i, \mathbf{w} \sim \text{Poisson}(\lambda_i), \quad \text{where } \lambda_i = \exp(\mathbf{w}^\top \mathbf{x}_i)$$

Recall that the Poisson distribution is defined as:

$$\text{Poisson}(y \mid \lambda) = \frac{e^{-\lambda} \lambda^y}{y!}, \quad y \in \mathbb{N}, \lambda > 0$$

where λ is the rate parameter of the distribution. The Poisson distribution is often used to model count data, it's defined for non-negative integers, and it has a mean equal to its variance. Assume the prior over the weights is Gaussian, $\mathbf{w} \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I})$.

- (3) (a) Write the log-likelihood function $p(\mathbf{y} \mid \mathbf{X}, \mathbf{w})$ under the model above, where $\mathbf{y} = (y_1, \dots, y_n)^\top$ and $\mathbf{X} = (\mathbf{x}_1, \dots, \mathbf{x}_n)^\top$ is the design matrix. You can assume that the observations are independent given the parameters.

- (3) (b) According to the probabilistic interpretation of loss minimization, minimizing a regularized loss is equivalent to what? Justify your answer.
- ☐ Maximization of the joint between likelihood and a prior induced by the regularization.
 - ☐ Maximization of the likelihood function.
 - ☐ Maximization of the marginal likelihood of the model.
 - ☐ The two quantities are not related.

- (5) (c) Write the log-posterior distribution $p(\mathbf{w} \mid \mathbf{y}, \mathbf{X})$ of the model above. You can ignore the constants that do not depend on $\mathbf{w}$.

- (5) (d) This model does not have a closed-form solution, so we need to use approximations to compute the posterior. What is the main idea behind Laplace approximation? Describe the steps you would take to compute the posterior using this method.

- (3) (e) What is the main problem with Laplace approximation? Motivate your answer.
- ☐ It's not a Bayesian method.
 - ☐ It captures global properties of the posterior, but not local ones.
 - ☐ It's not applicable to the Poisson regression model because the posterior is not Gaussian.
 - ☐ It computationally and memory expensive.

- (5) (f) You implemented the Laplace approximation and you now have an approximation $q(\mathbf{w})$ of the posterior $p(\mathbf{w} \mid \mathbf{y}, \mathbf{X})$. You want to compute the posterior predictive distribution of a new observation y_* given the new input $\mathbf{x}_*$. Write down the equation for this distribution. How would you do it? What is the main problem with this approach? How can you solve it?

- (5) (g) You also want to try a different approximation method and you decide for a sampling-based method. Describe the steps you would take to approximate the posterior using a Markov Chain Monte Carlo (MCMC) method with random walk Metropolis-Hastings. Highlight the main steps and the key properties of the method.

- (3) (h) Which of the following statements best describes a key property required for MCMC methods to produce valid samples from a posterior distribution? Motivate your answer.
- ☐ The proposal distribution must be symmetric and Gaussian.
 - ☐ The initial state of the Markov chain must be drawn from the true posterior.
 - ☐ The Markov chain must have sufficient mixing to explore the full posterior support.
 - ☐ The likelihood function must be conjugate to the prior distribution.

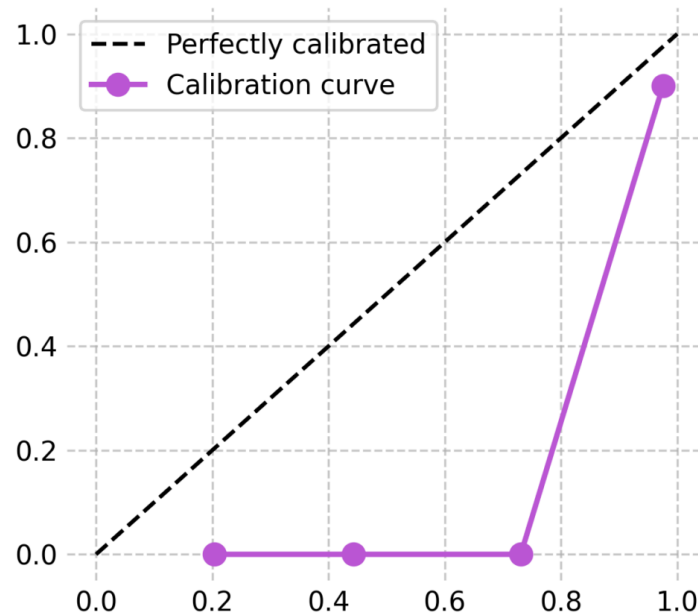
- (5) (i) You implemented the MCMC method and you have a set of samples $\mathbf{w}_1, \dots, \mathbf{w}_N$ from the posterior distribution. How can you be sure that these samples are valid? What tool can you use to check this?

3. **Deep Learning.** You are an engineer at an automotive company and you are working on a project to develop a new computer vision system for autonomous driving. You are tasked with designing a system that can accurately classify objects in real-time using a camera mounted on the vehicle. You decide to use a deep learning approach to solve this problem.

- (3) (a) What kind of likelihood would you use to model the data in this case? Justify your answer.
- ☐ Gaussian likelihood
 - ☐ Poisson likelihood
 - ☐ Categorical likelihood
 - ☐ Exponential likelihood
 - ☐ Bernoulli likelihood

- (5) (b) After training your model, you realize that it is overfitting the training data. You decide to regularize your model to improve its generalization performance. Describe two different regularization techniques you could use. Then, for one of them, explain what probabilistic interpretation it has.

- (5) (c) You want to be able to trust your model and you decide to evaluate its calibration. To assess whether your model's predicted probabilities are well-calibrated, you generate a reliability diagram (included below).



Describe what calibration means in this context. What are the axes of the plot? What does the plot tell you about the calibration of your model? How can you improve it?

- (5) (d) You decide to incorporate model uncertainty explicitly using Monte Carlo Dropout during inference. Describe the procedure and write the approximate predictive distribution $p(y \mid \mathbf{x}, \mathcal{D})$. Discuss its difference with respect to standard deterministic DNN inference.

4. **Generative models.** You are working on biomedical data and you are tasked with developing a generative model to synthesize new samples. You start simple and use a Gaussian Mixture Model (GMM) to model the data. You assume that the data is generated from a mixture of K Gaussian distributions, each with its own mean and covariance matrix.

- (5) (a) Write down the joint distribution of the data and the latent variables in a GMM model $p(\mathbf{x}, \mathbf{z})$, and a possible factorization.

- (3) (b) You want to learn the parameters of the Gaussian Mixture Model from observed data. What kind of optimization algorithm is typically used for this type of latent-variable model?
- ☐ Stochastic Gradient Descent
 - ☐ Newton-Raphson Optimization
 - ☐ Expectation-Maximization (EM)
 - ☐ Variational Inference

- (5) (c) You implemented the EM algorithm to fit a Gaussian Mixture Model on your biomedical dataset. After several iterations, the log-likelihood stops improving and the algorithm seems to converge. However, you are concerned that the solution may not be globally optimal. Explain why this can happen and what strategies can be used to address these issues in practice.

- (10) (d) You are analyzing gene expression data, where each observation $\mathbf{x}_i \in \mathbb{R}^2$ represents normalized measurements of two genes. You assume the data is generated from a mixture of $K = 2$ Gaussian components, each with a diagonal covariance matrix. The initial parameters of the model are:

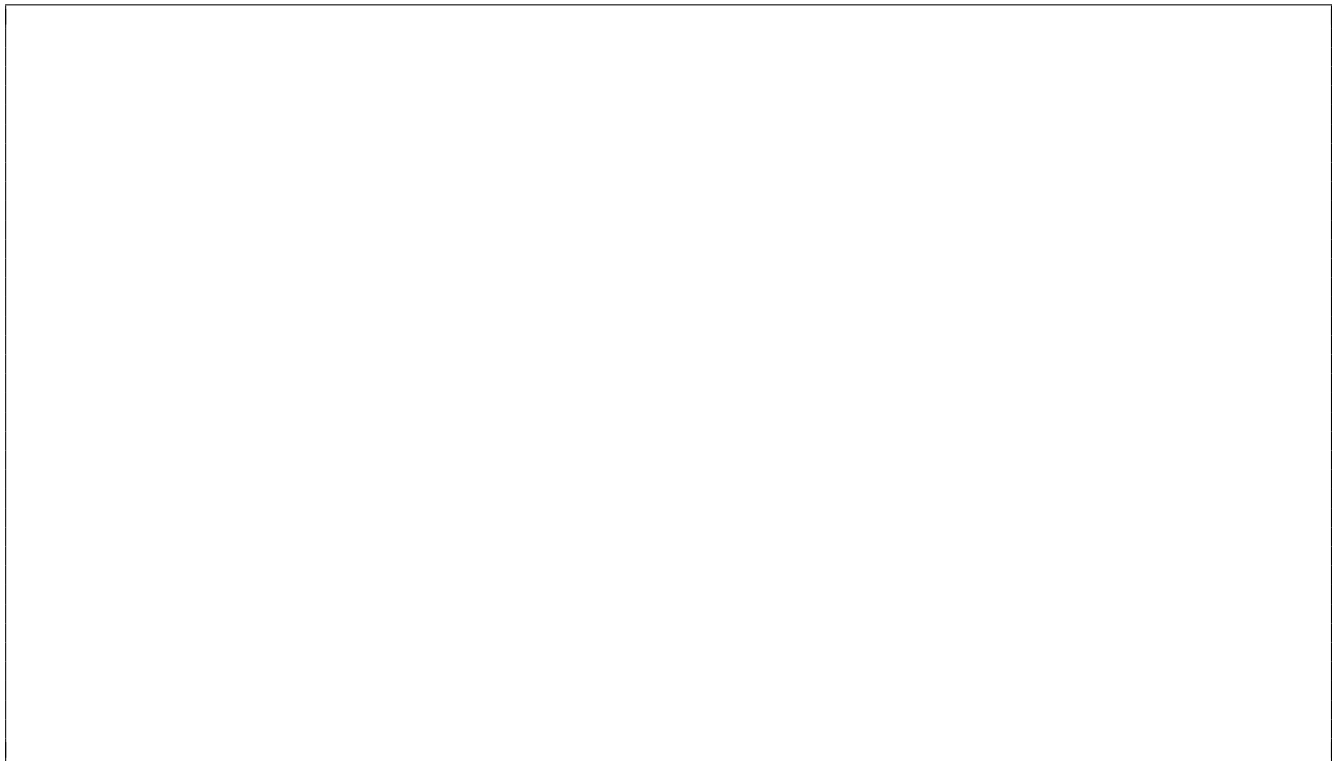
$$\pi_1 = \pi_2 = 0.5, \quad \boldsymbol{\mu}_1 = \begin{bmatrix} -1 \\ -1 \end{bmatrix}, \quad \boldsymbol{\mu}_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad \boldsymbol{\Sigma}_1 = \boldsymbol{\Sigma}_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

You observe a single data point:

$$\mathbf{x}_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Tasks (a figure may help):

1. Compute the responsibilities $\gamma(z_1 = k) = p(z_1 = k \mid \mathbf{x}_1)$ for $k = 1, 2$.
2. Explain the intuition behind your result. Which component is more likely to have generated the observation, or are they equally likely? Why?
3. Discuss what would happen if you observed more points near the origin. How would the optimization algorithm update the component means $\boldsymbol{\mu}_1$ and $\boldsymbol{\mu}_2$? Where would they end up after many iterations?



List of identities and properties

Here is a minimal list of identities and probability distributions strictly necessary to solve the exam. All other information can be derived or is standard background.

- **Binomial distribution:**

$$p(x \mid \theta) = \binom{n}{x} \theta^x (1 - \theta)^{n-x}, \quad x = 0, 1, \dots, n, \quad 0 < \theta < 1$$

- **Beta distribution:**

$$\text{Beta}(\theta \mid \alpha, \beta) = \frac{\theta^{\alpha-1} (1 - \theta)^{\beta-1}}{B(\alpha, \beta)}, \quad B(\alpha, \beta) = \frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha + \beta)}$$

- **Poisson distribution:**

$$p(y \mid \lambda) = \frac{e^{-\lambda} \lambda^y}{y!}, \quad y \in \mathbb{N}, \quad \lambda > 0$$

- **Gaussian distribution (univariate):**

$$\mathcal{N}(x \mid \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

- **Gaussian distribution (multivariate):**

$$\mathcal{N}(\mathbf{x} \mid \boldsymbol{\mu}, \boldsymbol{\Sigma}) = \frac{1}{(2\pi)^{d/2} |\boldsymbol{\Sigma}|^{1/2}} \exp \left(-\frac{1}{2} (\mathbf{x} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1} (\mathbf{x} - \boldsymbol{\mu}) \right)$$

- **Categorical distribution:**

$$p(y = k) = \pi_k, \quad \sum_{k=1}^K \pi_k = 1$$